

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

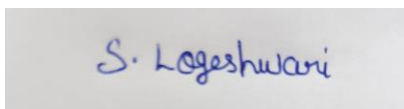
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I **Logeshwari S (192565040)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student

A rectangular box containing a handwritten signature in blue ink that reads "S. Logeshwari".

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message

Event Communication Security: Protecting Confidential Event Information Using the Caesar Cipher

Introduction

In today's digital world, secure communication plays an important role in protecting confidential information. Whether it is an organization, educational institution, business, or student club, sensitive information should only be accessible to authorized members. One of the simplest methods of protecting messages is encryption.

Encryption is the process of converting readable information (plaintext) into an unreadable format (ciphertext) using a specific algorithm or key. Only someone who knows the correct key can convert the encrypted message back into its original form through a process called decryption.

Before modern encryption techniques such as AES and RSA were developed, people relied on classical cryptographic methods. One of the earliest and most famous techniques is the **Caesar Cipher**, named after Julius Caesar, who reportedly used it to protect military communications.

In this assignment, we examine a scenario in which a student organization wants to securely communicate confidential event details to its core members using a Caesar Cipher with a shift of four positions. The assignment demonstrates both the encryption and decryption processes and discusses the advantages and limitations of this classical encryption method.

Objectives

The objectives of this assignment are:

- To understand the concept of classical cryptography.
- To study the Caesar Cipher encryption technique.
- To encrypt a message using a shift value of 4.
- To decrypt the encrypted message back to its original form.
- To understand the practical applications, advantages, and limitations of the Caesar Cipher.

What is Cryptography?

Cryptography is the science of securing information by transforming it into a form that unauthorized users cannot understand. It ensures that information remains confidential during transmission or storage.

The primary goals of cryptography include:

1. **Confidentiality**

- Prevents unauthorized users from reading sensitive information.

2. **Integrity**

- Ensures that the information has not been altered.

3. **Authentication**

- Verifies the identity of the sender.

4. **Non-Repudiation**

- Prevents the sender from denying that they sent the message.

Cryptography is widely used in:

- Online banking
- Email security
- Digital signatures
- Password protection
- Secure messaging applications
- Wireless communication
- Student and organizational communication

Classical Encryption Techniques : Classical cryptography consists of traditional encryption methods developed before the age of computers.

Some common classical encryption techniques include:

- Caesar Cipher
- Monoalphabetic Cipher
- Playfair Cipher
- Hill Cipher
- Vigenère Cipher
- Rail Fence Cipher

Among these, the Caesar Cipher is the simplest and easiest to understand.

Caesar Cipher

The Caesar Cipher is a substitution cipher where every letter in the plaintext is replaced by another letter located a fixed number of positions away in the alphabet.

For example:

Shift = 4

A → E

B → F

C → G

D → H

...

W → A

X → B

Y → C

Z → D

This shifting process is called encryption.

To recover the original message, the receiver shifts each letter backward by the same number.

Scenario Description

A student organization wants to securely share confidential event information with its core committee members.

Instead of sending the original message directly, they decide to encrypt the message using a Caesar Cipher with a fixed shift of four positions.

Original Message:

MEET AT NOON

Shift Value:

4

Only members who know the shift value can decode the message.

Part (a): Encryption Process

Plaintext

MEET AT NOON

Remove spaces for easier encryption:

MEETATNOON

Now encrypt every letter.

Plain Letter	Shift +4	Cipher Letter
M	Q	Q
E	I	I
E	I	I
T	X	X
A	E	E
T	X	X
N	R	R
O	S	S
O	S	S
N	R	R

Now insert spaces again.

Encrypted Message

QIIX EX RSSR

Therefore,

Plaintext:

MEET AT NOON

Ciphertext: QIIX EX RSSR

Step-by-Step Encryption Explanation

Letter M

$$M + 4 = Q$$

Letter E

$$E + 4 = I$$

Second E

$$E + 4 = I$$

Letter T

$$T + 4 = X$$

Space remains unchanged.

Letter A

$$A + 4 = E$$

Letter T

$$T + 4 = X$$

Space remains unchanged.

Letter N

$$N + 4 = R$$

Letter O

$$O + 4 = S$$

Second O

$$O + 4 = S$$

Letter N

$$N + 4 = R$$

Final encrypted message:

QIIX EX RSSR

Part (b): Decryption Process

An authorized member knows that the message was encrypted using a shift of four.

Therefore, every encrypted letter is shifted backward by four positions.

Encrypted Message:

QIIX EX RSSR

Cipher Letter	Shift -4	Plain Letter
Q	M	M
I	E	E
I	E	E
X	T	T
E	A	A
X	T	T
R	N	N
S	O	O
S	O	O
R	N	N

Recovered Message:

MEET AT NOON

Thus, the authorized member successfully retrieves the original confidential message.

Encryption and Decryption Formula

Encryption Formula:

$$C = (P + K) \bmod 26$$

Where:

- C = Ciphertext
- P = Plaintext
- K = Shift Key
- K = 4

Decryption Formula:

$$P = (C - K) \bmod 26$$

Where:

- P = Plaintext
- C = Ciphertext

- K = Shift Key

Working Example

Suppose the message is:

HELLO

Shift = 4

Encryption:

H $\rightarrow$ L

E $\rightarrow$ I

L $\rightarrow$ P

L $\rightarrow$ P

O $\rightarrow$ S

Encrypted text:

LIPPS

Decryption:

L $\rightarrow$ H

I $\rightarrow$ E

P $\rightarrow$ L

P $\rightarrow$ L

S $\rightarrow$ O

Recovered text:

HELLO

This demonstrates how encryption and decryption are inverse operations.

Advantages of Caesar Cipher

1. Very easy to understand.
2. Simple to implement manually.
3. Requires very little computational power.
4. Useful for learning basic cryptography.
5. Demonstrates the concepts of encryption and decryption clearly.
6. Suitable for educational purposes.

Limitations of Caesar Cipher

Despite its simplicity, the Caesar Cipher has many weaknesses.

1. Small Key Space

Only 25 possible keys exist.

An attacker can easily try every key.

2. Vulnerable to Brute Force Attack

Since there are only 25 shifts, attackers can quickly discover the correct key.

3. Frequency Analysis

Letters such as E, T, and A occur frequently in English.

Attackers can use frequency analysis to guess the encryption.

4. Not Suitable for Modern Security

Modern communication requires stronger algorithms like AES, RSA, and ECC.

Applications of Classical Encryption

Although outdated for serious security, classical encryption methods are still useful in:

- Educational demonstrations
- Cryptography laboratories
- Puzzle games
- Treasure hunts
- Coding competitions
- Secret classroom activities
- Historical research

Real-World Importance of Secure Communication

Student organizations often communicate confidential information such as:

- Meeting schedules
- Event locations
- Budget details
- Volunteer assignments
- Examination-related notices
- Competition strategies

Even though the Caesar Cipher is not sufficiently secure for modern communication, it introduces the basic principles of protecting sensitive information before more advanced encryption techniques are learned.

Comparison Between Plaintext and Ciphertext

Plaintext	Ciphertext
MEET	QIIX
AT	EX
NOON	RSSR

Complete Message:

Plaintext:

MEET AT NOON

Ciphertext:

QIIX EX RSSR

Conclusion

Cryptography is essential for protecting confidential information. The Caesar Cipher is one of the earliest and simplest encryption methods and serves as an excellent introduction to the concepts of secure communication.

In this assignment, the original message "**MEET AT NOON**" was encrypted using a shift value of **4**, resulting in the ciphertext "**QIIX EX RSSR**." An authorized member who knows the shift value can decrypt the message by shifting each letter four positions backward, successfully recovering the original message.

Although the Caesar Cipher is not secure enough for modern applications due to its limited key space and vulnerability to brute-force attacks, it remains valuable for educational purposes. It helps students understand the fundamental ideas behind encryption, decryption, keys, plaintext, and ciphertext, which form the foundation of modern cryptographic systems. Overall, this exercise demonstrates the importance of secure communication and provides a practical understanding of classical encryption techniques used in information security.

2. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality. a. Demonstrate how the encryption process transforms the message step-by-step. b. Explain how the repeating keyword influences each character of the ciphertext.

Secure Messaging System Using Keyword-Based Encryption (Vigenère Cipher)

Introduction

In today's digital era, organizations exchange a large amount of confidential information through emails, messaging platforms, and internal communication systems. Protecting this information from unauthorized access is essential for maintaining privacy, security, and business continuity. Cryptography provides the necessary techniques to secure communication by converting readable messages into coded text that can only be understood by authorized users.

One of the classical encryption techniques used to enhance message confidentiality is the **Vigenère Cipher**, a keyword-based substitution cipher. Unlike the Caesar Cipher, which uses a single fixed shift for every letter, the Vigenère Cipher uses a repeating keyword to generate different shifts for different letters. This makes the encryption process more secure and difficult to break.

In this assignment, a company uses the keyword "**KEY**" to encrypt the message "**HELLO**" before sending it to an employee. The assignment demonstrates the complete encryption process, explains how the repeating keyword influences each character of the ciphertext, and discusses the advantages, limitations, and applications of the Vigenère Cipher.

Objectives

The objectives of this assignment are:

- To understand the concept of keyword-based encryption.
- To study the Vigenère Cipher technique.
- To perform encryption of the message **HELLO** using the keyword **KEY**.
- To understand how the repeating keyword affects each character during encryption.
- To learn the advantages, limitations, and applications of the Vigenère Cipher.

What is Cryptography?

Cryptography is the science of protecting information by converting readable data into an unreadable format. It ensures that only authorized individuals can access confidential information.

The primary goals of cryptography are:

1. **Confidentiality** – Prevents unauthorized users from reading the message.
2. **Integrity** – Ensures the message is not altered during transmission.
3. **Authentication** – Confirms the identity of the sender.
4. **Non-Repudiation** – Prevents the sender from denying that the message was sent.

Cryptography is widely used in:

- Secure email communication
- Online banking
- Digital payments
- Password protection
- Secure messaging applications
- Government communication
- Military communication
- Corporate information systems

Keyword-Based Encryption (Vigenère Cipher)

The Vigenère Cipher is a classical polyalphabetic substitution cipher that encrypts text using a repeating keyword. Each letter of the keyword determines how many positions the corresponding plaintext letter is shifted in the alphabet.

Unlike the Caesar Cipher, which uses the same shift throughout the message, the Vigenère Cipher changes the shift depending on the keyword, making it more secure.

For example:

Keyword = **KEY**

The corresponding shift values are:

- $K = 10$
- $E = 4$
- $Y = 24$

These shifts repeat until the entire message is encrypted.

Scenario Description

A company wants to secure its internal communication using keyword-based encryption.

The employee sends the message:

HELLO

using the keyword:

KEY

Since the keyword is shorter than the message, it is repeated until it matches the length of the plaintext.

Plaintext:

HELLO

Repeated Keyword:

KEYKE

Each letter of the keyword determines the shift used for encrypting the corresponding letter of the plaintext.

Part (a): Step-by-Step Encryption Process

Step 1: Write the Plaintext

Plaintext:

HELLO

Step 2: Repeat the Keyword

Keyword:

KEY

Repeated Keyword:

KEYKE

Now both strings have the same length.

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E

Step 3: Convert Letters into Alphabet Positions

Using A = 0, B = 1, C = 2, ..., Z = 25:

Letter	Position
H	7
E	4
L	11
L	11
O	14

Keyword positions:

Letter	Position
K	10
E	4

Letter	Position
Y	24
K	10
E	4

Step 4: Apply the Encryption Formula

The encryption formula is:

$$\text{Ciphertext} = (\text{Plaintext} + \text{Keyword}) \bmod 26$$

Now encrypt each letter:

Letter 1

$$H = 7$$

$$K = 10$$

$$7 + 10 = 17$$

$$17 = R$$

Letter 2

$$E = 4$$

$$E = 4$$

$$4 + 4 = 8$$

$$8 = I$$

Letter 3

$$L = 11$$

$$Y = 24$$

$$11 + 24 = 35$$

$$35 \bmod 26 = 9$$

$$9 = J$$

Letter 4

$$L = 11$$

$$K = 10$$

$$11 + 10 = 21$$

$$21 = V$$

Letter 5

$$O = 14$$

$$E = 4$$

$$14 + 4 = 18$$

$$18 = S$$

Final Ciphertext

Encrypted Message:

RIJVS

Therefore,

Plaintext:

HELLO

Keyword:

KEY

Ciphertext:

RIJVS

Encryption Table

Plaintext Letter	H	E	L	L	O
Plaintext Value	7	4	11	11	14
Keyword Letter	K	E	Y	K	E
Keyword Value	10	4	24	10	4
Sum (mod 26)	17	8	9	21	18
Ciphertext Letter	R	I	J	V	S

Final encrypted text:

RIJVS

Part (b): Influence of the Repeating Keyword

The keyword **KEY** is repeated as **KEYKE** because the message is longer than the keyword.

Each letter of the keyword creates a different shift value:

- K shifts the plaintext by **10** positions.
- E shifts the plaintext by **4** positions.
- Y shifts the plaintext by **24** positions.
- K again shifts by **10** positions.
- E again shifts by **4** positions.

Because different letters use different shifts, the same plaintext letter may produce different ciphertext letters depending on the corresponding keyword letter. This makes the Vigenère Cipher much stronger than a simple Caesar Cipher, where every letter is shifted by the same amount.

The repeating keyword introduces variation into the encryption process, making frequency analysis more difficult and increasing the overall security of the encrypted message.

Decryption Process

An authorized employee who knows the keyword **KEY** can recover the original message using the decryption formula:

$$\text{Plaintext} = (\text{Ciphertext} - \text{Keyword} + 26) \bmod 26$$

Applying this formula:

Ciphertext	R	I	J	V	S
Keyword	K	E	Y	K	E
Plaintext	H	E	L	L	O

Recovered message:

HELLO

Thus, only users with the correct keyword can decrypt and read the original message.

Advantages of the Vigenère Cipher

The Vigenère Cipher offers several advantages over simpler substitution ciphers:

- Uses multiple shift values instead of a single shift.
- More secure than the Caesar Cipher.
- Difficult to crack using simple frequency analysis.
- Easy to implement manually and programmatically.
- Provides a strong foundation for understanding modern encryption techniques.

Limitations of the Vigenère Cipher

Despite its improvements, the Vigenère Cipher has some limitations:

- If the keyword is short, patterns may repeat.

- Long messages with repeated keywords can be vulnerable to cryptanalysis.
- Modern computers can break the cipher using advanced analytical techniques.
- It is not suitable for protecting highly sensitive information in modern communication systems.

Applications of the Vigenère Cipher

Although it is no longer used for high-security applications, the Vigenère Cipher remains valuable in:

- Cryptography education
- Information security laboratories
- Puzzle competitions
- Coding challenges
- Historical research
- Learning classical encryption techniques

Comparison Between Plaintext and Ciphertext

Plaintext	HELLO
Keyword	KEYKE
Ciphertext	RIJVS

This example demonstrates how the same keyword encrypts each character differently, producing a secure encoded message.

Conclusion

Secure communication is an essential requirement for organizations to protect confidential information from unauthorized access. The Vigenère Cipher is a classical keyword-based encryption technique that improves security by using a repeating keyword to generate different shift values for different letters.

In this assignment, the plaintext "**HELLO**" was encrypted using the keyword "**KEY**." The keyword was repeated as "**KEYKE**", and each letter influenced the encryption of the corresponding plaintext character.

The resulting ciphertext was "**RIJVS.**" The authorized receiver, knowing the same keyword, can decrypt the message by reversing the encryption process and recover the original text "**HELLO.**"

Although modern encryption algorithms such as AES and RSA provide much stronger security, the Vigenère Cipher remains an important learning tool because it introduces the concepts of keyword-based encryption, polyalphabetic substitution, encryption keys, and secure communication. Understanding these classical methods helps build a strong foundation for studying advanced cryptographic techniques used in today's digital world.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation